

Trader Performance vs Market Sentiment Analysis

Data Science/Analytics Internship Assignment – Primetrade.ai

Methodology

The objective of this project was to analyze the relationship between Bitcoin market sentiment and trader performance on the Hyperliquid platform. Two datasets were used: the Bitcoin Fear & Greed Index and the Hyperliquid historical trader dataset.

The analysis began with loading both datasets into Google Colab using Python. Data preprocessing included checking for missing values and duplicate records, converting timestamps into a standard date format, and creating a common date column. The two datasets were then merged based on the date to associate each trading record with the corresponding market sentiment.

After merging, exploratory data analysis (EDA) was performed by calculating key metrics such as daily profit and loss (PnL), average PnL, total PnL, trade frequency, and the distribution of trades across different market sentiment categories. Visualizations including bar charts and trend plots were created to compare trading performance under different market conditions.

Key Insights

- Most trading activity occurred during **Fear** market conditions, indicating higher trader participation during periods of uncertainty.
- Trading performance (Closed PnL) differed across market sentiment categories, suggesting that sentiment has an impact on profitability.
- The merged dataset provided a clear view of how trader behavior changes with market sentiment.
- Visualizations showed noticeable differences in profit trends between Fear, Greed, Neutral, and Extreme Greed periods.
- Market sentiment can be a valuable indicator when evaluating trading strategies and risk.

Strategy Recommendations

- Reduce leverage and manage risk more carefully during **Fear** periods because market volatility tends to be higher.
- Consider increasing trading activity during **Greed** periods only when supported by additional technical or market indicators.
- Use the Fear & Greed Index as a supporting signal rather than relying on it alone for trading decisions.

- Continuously monitor market sentiment along with trade performance to improve risk management and decision-making.

Conclusion

This analysis demonstrates that market sentiment influences both trader behavior and trading performance. Integrating sentiment indicators with trading metrics can help traders make more informed decisions and develop more effective trading strategies. The project highlights the importance of combining data analysis with market psychology to improve trading outcomes.